Ignytion io

IGNYTION IO · TECHNICAL LAB NOTE

From SPICE to GDS: a SKY130 Amplifier

Two transistors, an external bias, and a checked physical implementation

An inverter can amplify a small analogue signal when its input is biased near the transition region. This note follows a SKY130 CMOS inverter from transistor simulation through layout, design-rule checking, schematic equivalence, and capacitance extraction. It reports the measured gain and the dependencies needed to repeat this local experiment.

SECTION 01

What this experiment establishes

A **1 mV peak input at 1 kHz** produces an inverted output of approximately 135 mV peak. Both the schematic and extracted circuit pass the declared numerical checks.

CHECK	OBSERVED RESULT	STATE
Schematic AC gain	135.86 V/V at 1 kHz	✓ PASS
Extracted AC gain	135.86 V/V at 1 kHz	✓ PASS
Magic DRC	0 reported errors	✓ PASS
Netgen LVS	2 devices · 4 nets · unique match	✓ PASS
Negative controls	Open gate and corrupt waves rejected	✓ PASS

> **Read the scope before using the numbers**
> This is a **TT, 27 °C, 1.8 V** educational experiment with a 100 fF ideal load. The extracted model includes capacitance. It excludes interconnect resistance, process variation, and fabricated-silicon measurements.

Crucible is beta. Its environment records the managed tools used here. A separately acquired Tcl-capable Magic build supplies the layout generator because the installed classic binary cannot execute the PDK scripts.

SECTION 02

How the amplifier works

The **PMOS** transistor pulls the output towards the supply. The **NMOS** pulls it towards ground. Both gates receive the same input.

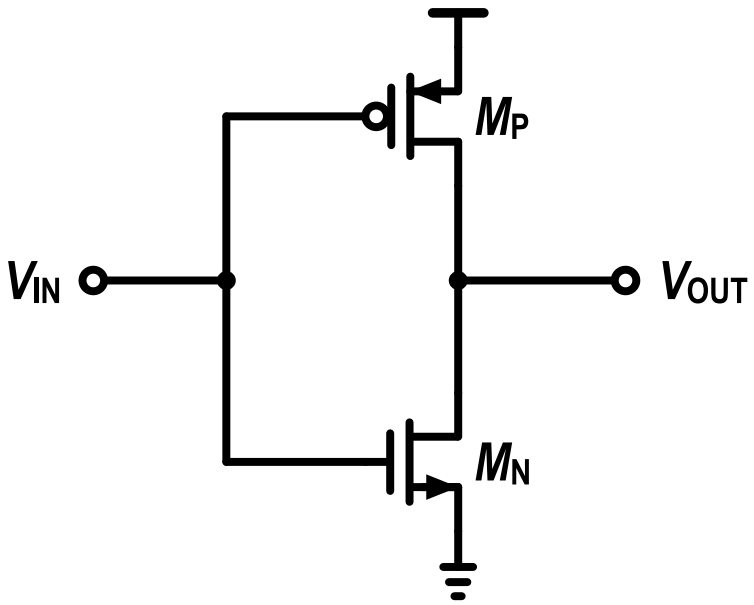


Figure 1 · CMOS inverter amplifier · body terminals tied to their source supplies

DEVICE	W / L	MODEL
MN	2 / 0.5 µm	SKY130 1.8 V NMOS
MP	6 / 0.5 µm	SKY130 1.8 V PMOS

Near the transition, a higher input strengthens the NMOS and weakens the PMOS. The output falls. The steep negative slope becomes **inverting voltage gain** for a sufficiently small signal.

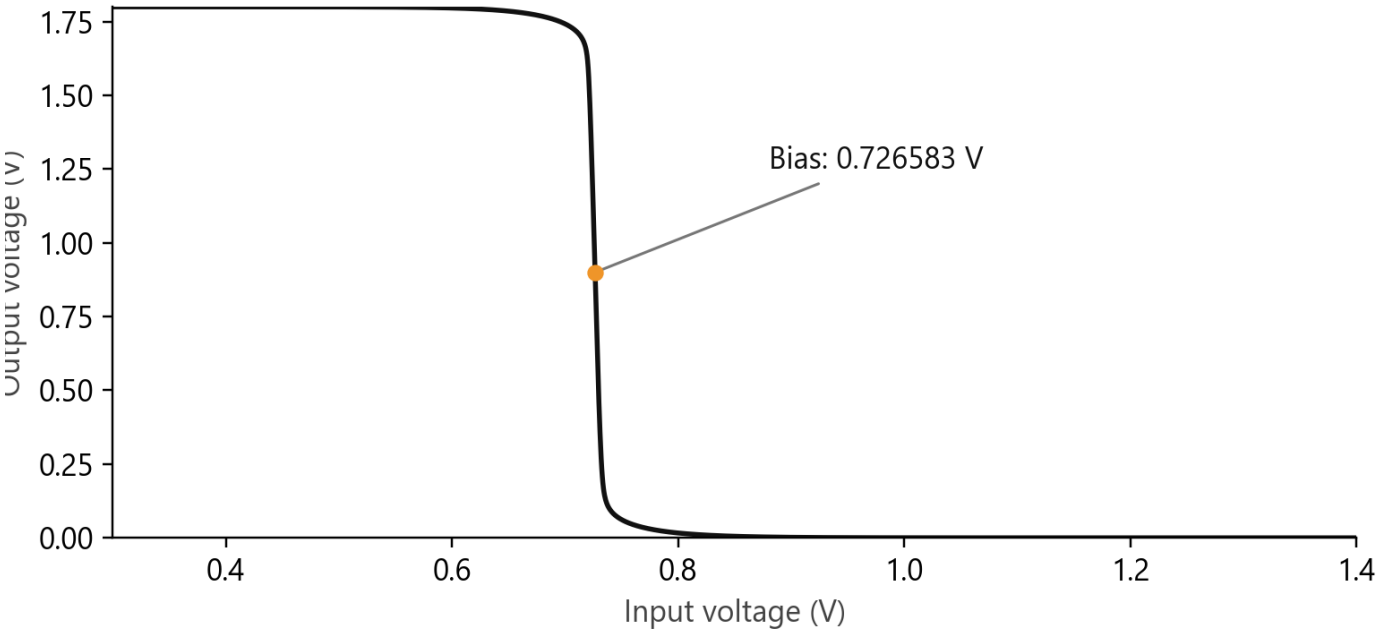
The source supplies are 1.8 V and ground. Ideal sources set the input bias; the bias circuit and 100 fF output load are outside the layout.

SECTION 03

Find the operating point

The **DC sweep** varies the input from 0.3 V to 1.4 V in 0.5 mV steps. Linear interpolation locates the input that places the output at half the supply.

DC TRANSFER · SCHEMATIC SIMULATION



QUANTITY	MEASURED VALUE
Selected input bias	0.726583174 V
Schematic output	0.899991742 V
Supply current	6.528463 µA

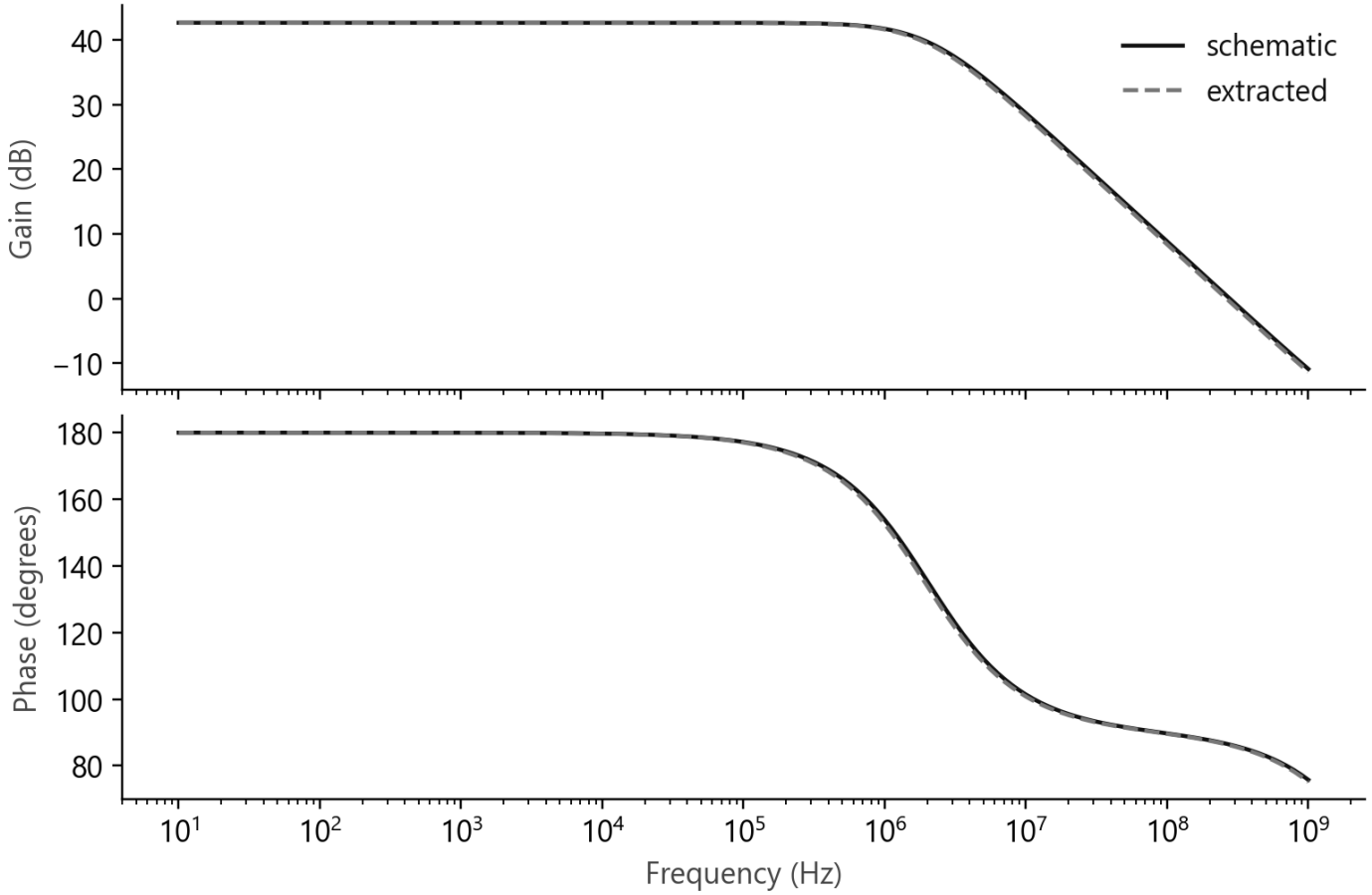
A separate operating-point simulation checks the interpolated bias. Its output is within the predeclared **20 mV** window around 0.9 V. The narrow transition makes external bias accuracy significant.

SECTION 04

Measure small-signal gain

AC analysis linearises the devices around the operating point. A unit AC input makes the output magnitude equal to the voltage-gain magnitude.

AC RESPONSE · SAME BIAS AND LOAD



At 1 kHz, the schematic gain is **135.86 V/V**. The extracted gain rounds to the same value. The phase is approximately 180°, consistent with inversion.

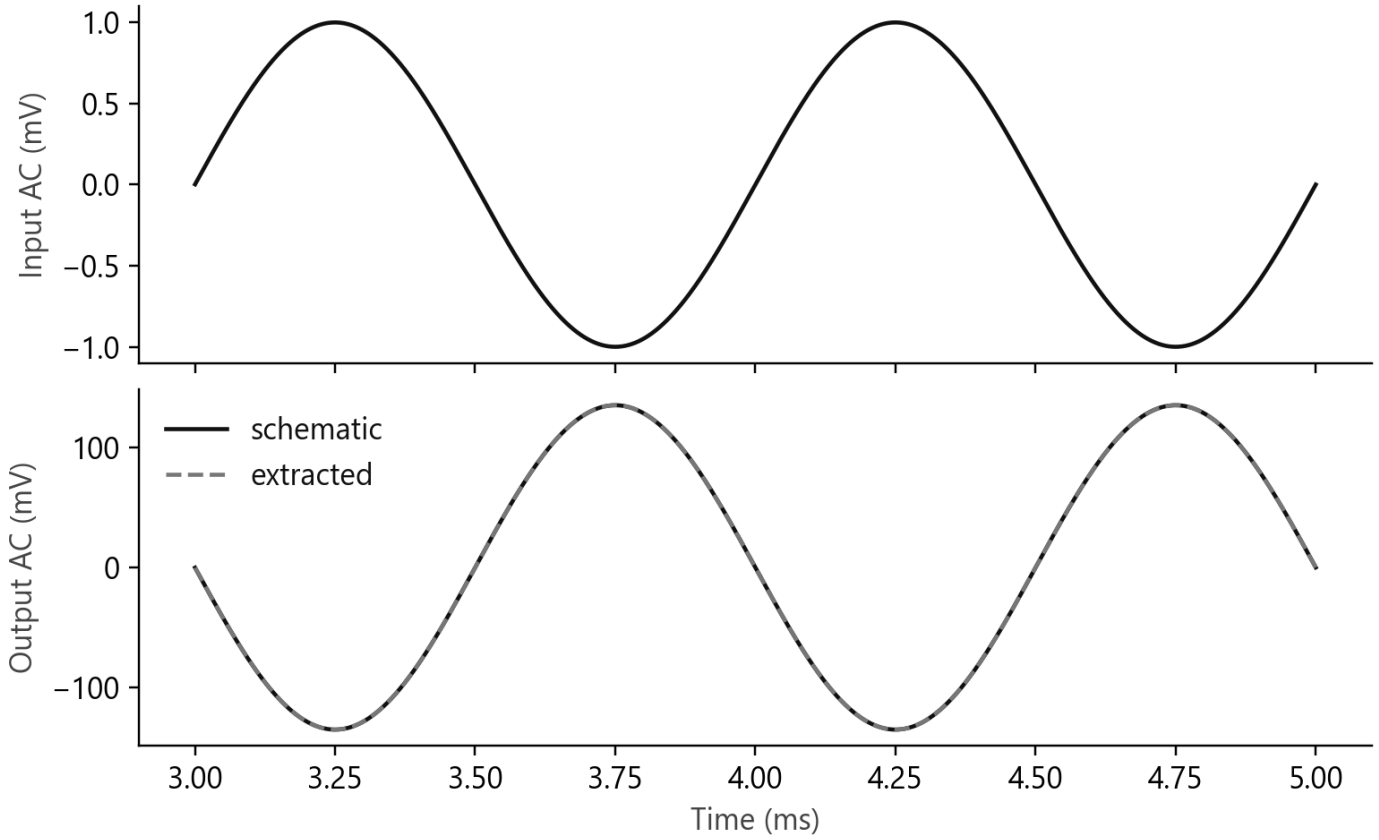
The curves nearly overlap at low frequency. The frequency sweep shows the capacitive roll-off; these results are specific to this circuit and load. No gain-bandwidth or stability guarantee is inferred for another application.

SECTION 05

Check a real waveform in simulation

Transient analysis applies a 1 mV peak sine wave at 1 kHz for 5 ms. A sine-and-cosine fit over the final 2 ms measures amplitude and polarity after settling.

TRANSIENT RESPONSE · FINAL TWO PERIODS



The signed schematic gain is **-135.463 V/V**, and the output fundamental is **135.463 mV peak**. The fitted gain differs from the AC magnitude by less than the declared 10% limit.

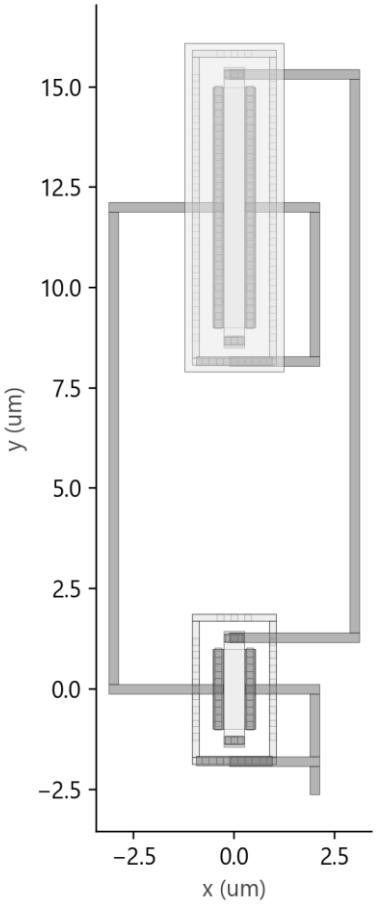
Flattening the output or reversing its polarity makes the same observer fail. This checks that completion of the simulator alone cannot produce a passing verdict.

SECTION 06

Connect the physical devices

The layout uses **native SKY130 device generators** with substrate and well contacts. Metal 1 joins the gates, joins the drains, and connects each source to its body supply.

DECODED GDS GEOMETRY · MICROMETRES



The overview is decoded from the actual GDS stream. It uses neutral layer shading for readability. Four ports identify input, output, supply, and ground; the testbench components are external.

The GDS contains **3 cells**. Its bounding box is approximately **6.24 × 18.72 µm**. This is a demonstration layout without pads, ESD structures, or a fabrication integration wrapper.

SECTION 07

Verify geometry and connectivity

Design-rule checking (DRC) checks layout geometry. **Layout versus schematic (LVS)** compares extracted connectivity and device properties against the schematic. Each needs its own evidence.

```
AMPLIFIER_DRC_COUNT 0
No errors found.
Final result: Circuits match uniquely.
Circuit 1 contains 2 devices, Circuit 2 contains 2 devices.
Circuit 1 contains 4 nets,    Circuit 2 contains 4 nets.
```

These are trimmed lines from the final Magic and Netgen logs. The Python checker also decodes the GDS and verifies the named top cell and non-empty geometry.

CONTROL	REQUIRED OBSERVATION
PMOS gate disconnected	LVS mismatch in the saved control report
Flat output waveform	Numerical observer rejects lost gain
Output polarity reversed	Numerical observer rejects lost inversion

Open-PDK checks have a defined boundary

The reported **zero DRC errors** refer to this Magic deck and this layout. A unique LVS match establishes this two-device circuit comparison. Neither result is foundry signoff or tapeout approval.

SECTION 08

Compare the extracted circuit

Magic extraction produces **five parasitic capacitors** between the circuit nets. The extracted transistor netlist also records device diffusion areas and perimeters.

METRIC	SCHEMATIC	EXTRACTED
Output bias	0.899992 V	0.899992 V
Gain at 1 kHz	135.857249 V/V	135.857247 V/V
Transient gain	-135.462722	-135.462718
Supply current	6.528463 µA	6.528463 µA

The low-frequency gain change is below the declared **20%** limit. The gain is reported to six decimals here to show the comparison, without implying that the transistor model predicts silicon to that precision.

This flow performs **capacitance extraction**. It does not run interconnect-resistance extraction. The frequency-domain difference is therefore a bounded capacitive comparison, and further extraction would be needed for an RC claim.

SECTION 09

Repeat the local experiment

Use the existing **WSL project workspace**. The shell scripts select the project environment explicitly. Their dependency paths are recorded rather than treated as portable installation defaults.

```
# Start inside the Analog_SPICE_to_GDS folder in WSL.
igny init
bash scripts/run.sh prepare
bash scripts/run.sh sim
bash scripts/run.sh layout > results/layout.log 2>&1
bash scripts/run.sh lvs > results/netgen.log 2>&1
bash scripts/run.sh characterise
```

On the Windows host, the independent artifact checker uses Python with NumPy and KLayout. These are **analysis dependencies**, separate from the managed simulator and LVS tools.

```
python scripts/verify.py simulation
AMPLIFIER_SIMULATION_VERIFIED_WITH_NEGATIVE_CONTROLS
python scripts/verify.py physical
AMPLIFIER_PHYSICAL_ARTIFACTS_VERIFIED
```

The scripts and raw tables are the reproduction record. A new machine also needs the pinned PDK, the Tcl-capable Magic payload, FFTW, and the host analysis libraries. Clean-machine reproduction has not been tested.

SECTION 10

Dependencies and evidence

DEPENDENCIES.json records file hashes and local provenance. The workspace lock records the three managed bindings. A lock entry for classic Magic does not pin the separate Tcl build used here.

DEPENDENCY	ACTUAL USE
ngspice 46	Managed transistor simulation
Netgen 1.5.323	Managed schematic comparison
Magic 8.3.682 with Tcl	Separately acquired layout/extraction payload
SKY130 revision	1689ac3f2dc7… · exact identity in manifest
FFTW runtime	Existing local library copied into build/runtime
NumPy and KLayout	Host numerical checks and GDS decoding

The active PDK lacked corner include files. The experiment uses another local revision containing the required models. It loads the NMOS and PMOS TT model families, their mismatch parameters with statistical switches disabled, and the supplied layout-dependent parameters.

A project-specific Magic startup file resolves the PDK paths and enables the finer grid expected by the generator. The first coarse-grid attempts failed DRC; the final layout was regenerated and checked at the correct grid.

EVIDENCE FILES

Raw sweep, AC, and transient tables sit in `results/`. The final GDS, editable Magic views, LVS netlist, and capacitive netlist sit in `build/`. Numerical and physical summaries are regenerated from those artifacts.

SECTION 11

Limits and source notes

The measured result is an **externally biased inverter amplifier at one operating condition**. It provides a small physical-verification example that can be inspected and rerun locally.

No process-corner sweep, temperature sweep, mismatch study, noise analysis, distortion specification, bias servo, resistive extraction, ESD integration, or fabricated-silicon measurement is covered. The apparent numerical precision of the tables must not be read as manufacturing accuracy.

SOURCE NOTES

SKY130 device documentation
`https://skywater-pdk.readthedocs.io/en/main/rules/device-details.html`

Open PDKs · SKY130 Magic generators
`https://github.com/fossi-foundation/open-pdks/blob/main/sky130/magic/sky130.tcl`

ngspice documentation
`https://ngspice.sourceforge.io/docs.html`

Magic documentation
`http://opencircuitdesign.com/magic/`

Netgen documentation
`http://opencircuitdesign.com/netgen/`

The figures and result tables in this note come from the local experiment dated **27 October 2026**. Tool projects and PDK authors retain credit for their work. No third-party endorsement is implied.

> **Next engineering step**
> A bias circuit and a PVT study would be needed before selecting this topology for a specified application. Those activities require their own requirements and evidence.